

Primitive recursive functions

basic functions + $\begin{cases} \text{composition} \\ \text{recursive definition} \end{cases}$

primitive recursive $\not\subseteq$ computable

μ -recursive functions

basic functions + $\begin{cases} \text{composition} \\ \text{recursive definition} \\ \text{minimization of } \text{minimizable functions} \end{cases}$

$\nearrow$ closed

μ recursive functions

minimizable functions
undecidable

primitive recursive $\subseteq \mu$ -recursive

Theorem

A numerical function f is μ -recursive iff it is computable \ recursive.

Proof:

μ -recursive $\Rightarrow$ computable

1. basic functions are computable
2. three operations preserve computability

computable $\Rightarrow \mu$ -recursive (Sketch)

$\exists M$ computes f

$(s, \triangleright \cup n) \vdash_M (q_1, \triangleright \cup q, v) \vdash_M \dots \vdash_M (h, \triangleright \cup f(n))$

$\Delta U S n$ $t_n \Delta u_i q_i v_i t_n \dots t_n \Delta u h f(n)$

$\Sigma \cup K$ to $\{0, 1, 2, \dots, b-1\}$ where $b = |\Sigma| + |K|$

base- b integer

$$h^*(\Delta U S n) = \Delta U S n \Delta u_i q_i v_i \Delta \dots \Delta u h f(n)$$

$$= \mu m [\text{iscomp}(\Delta U S n, m) \wedge \text{halted}(m)]$$

$$\text{iscomp}(\Delta U S n, m) = \begin{cases} 1 & \text{if } m \text{ represent a computation} \\ & \text{starting with } \Delta U S n. \\ 0 & \text{otherwise} \end{cases}$$

primitive
recursive

$$\text{halted}(m) = \begin{cases} 1 & \text{if } m \text{ ends with } \Delta u h f(n) \\ 0 & \text{otherwise.} \end{cases}$$

→ μ -recursive

f : on input n

Primitive recursive $\leftarrow h_0: n \longrightarrow \Delta U S n$

μ -recursive $\leftarrow h^*: \Delta U S n \longrightarrow \Delta U S n \Delta u_i q_i v_i \Delta \dots \Delta u h f(n)$

P.R $\leftarrow h_1: \Delta U S n \Delta \dots \Delta u h f(n) \longrightarrow \Delta u h f(n)$

P.R $\leftarrow h_2: \Delta u h f(n) \longrightarrow f(n)$

regular
expression

NFA

DFA $\xrightarrow{\text{accept}}$ regular language

pumping theorem

closure property:

U. o. * . $\cap$. -
 ✓ ✓ ✓ ✓ ✓

Chomsky
norm form

parse
tree

CFG

generate

CFL

pumping theorem

PDA

accept

closure
property

U. o. * . $\cap$. -
 ✓ ✓ ✓ x x

multitape

non-deterministic

Enumerator

Grammar

algorithm

if halts on
every input

TM

variants

numerical
function

μ -recursive

basic
functions

primitive
recursive

decide

semi
decide

compute

decidable \ recursive
language

recursively
enumerable
language

computable \ recursive
function

decidable
problem

not
in

H

in (semidecided by U)
complete

universal
TM

closure property

closure property

U. o. * . $\cap$. -

✓ ✓ ✓ ✓ ✓

U. o. * . $\cap$. -
 ✓ ✓ ✓ ✓ x

$\bar{H}$

reduction (A $\leq$ B)

more undecidable problem

Rice's theorem

L is recursive iff

L and $\bar{L}$ are recursively
enumerable

countable

regular $\subsetneq$ C.F $\subsetneq$ recursive $\subsetneq$ recursively enumerable $\subsetneq$ universal
set

$\{a^i b^j : i \geq 0\}$ $\{a^i b^j c^k : i \geq 0\}$ H

$\bar{H}$

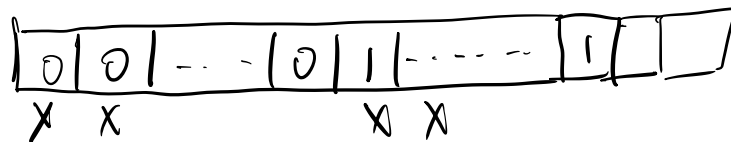
complexity theory

decidable vs undecidable

/ \
efficiently vs inefficient

resource: time, space

$$A = \{0^k 1^k : k \geq 0\}$$



$M_1 =$ on input w

$$n = |w|$$

1. scan across the tape, reject if a 0 appears to the right of a 1.

2. while there are both 0 and 1 on the tape

scan across the tape

cross a single 0 and a single 1

3. If neither 0 nor 1 remain on the tape

accept

else

reject

$n^2 + 3n$ steps

$$\begin{aligned} & \frac{1}{2} \cdot 2n \\ & \sim \frac{n}{2} \cdot 2n = n^2 \end{aligned}$$

n

Definition

Let M be a deterministic TM that halts on every input.
The running time of M is the function $f: \mathbb{N} \rightarrow \mathbb{N}$,
where $f(n)$ is the maximum steps that M uses on any input
of length n .

M runs in time $f(n)$

M is a $f(n)$ time TM.

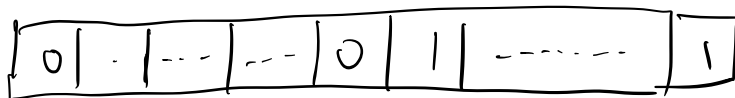
M_1 runs in $\overset{\text{time}}{\sqrt{n^2+3n}} \quad O(n^2)$

Definition

Let $t: \mathbb{N} \rightarrow \mathbb{R}^+$ be a function. Define the time complexity class
 $\text{DTIME}(t(n))$ be collection of all languages that are decidable
by $O(t(n))$ -time deterministic single-tape TM.

$A \in \text{DTIME}(n^2)$

$A \in \text{DTIME}(n \log n)$



$M_2 =$ on input w

1. scan across the tape, reject if "
2. repeat if both 0 and 1 remain on the tape.

$\log n$
 $4n$

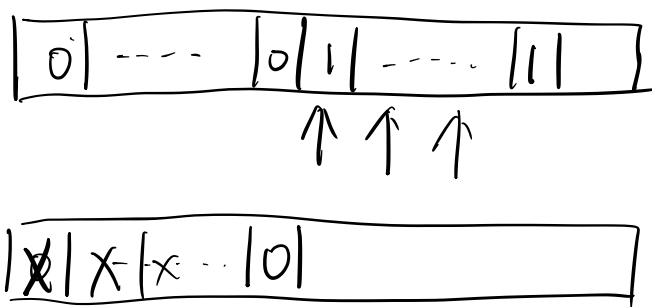
3. Scan across the tape, reject if $\#0 + \#1$ is odd.
Scan across the tape, cross every other 0
and every other 1

4. If neither 0 nor 1 remain on the tape
accept
else
reject

M_2 runs in time $O(n \log n)$

$A \in DTIME(n \log n)$

Two-tape M_3

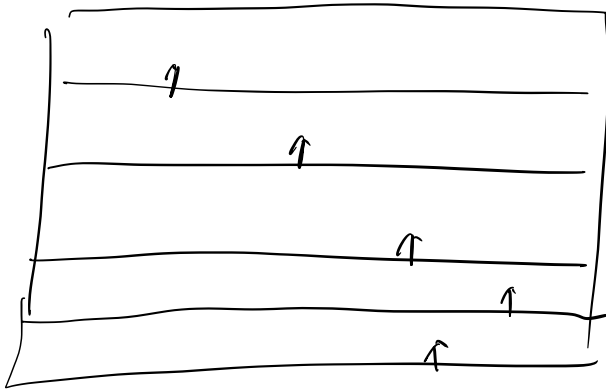


M_3 runs in time $O(n)$

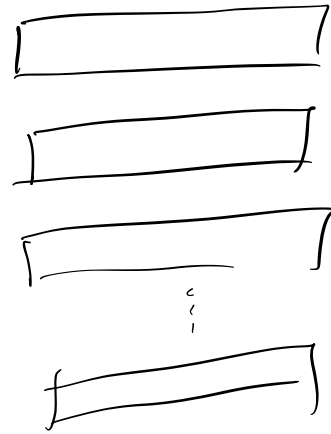
Theorem

Let $t(n)$ be a function, where $t(n) \geq n$. Then every $t(n)$ -time multitape TM has an equivalent $O(t^2(n))$ -time single tape Turing machine.

single-tape



multitape



scan twice : $4 \cdot t(n)$ steps

1 steps

total : $4 \cdot t^2(n)$ steps

total : $t(n)$

$O(t^2(n))$

Definition

A language is polynomially decidable if it is decided by some deterministic TM with running time $\text{poly}(n)$, where $\text{poly}(n)$ is a polynomial of n .

The time complexity class P is the set of all polynomially decidable language.

Corollary

$$P = \bigcup \text{DTIME}(n^k)$$

$k \geq 1$

Theorem

Every context-free language is in P.